

SURGE-090 · Smart Snake Robot

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Status: 25% Milestones Achieved

Executive Summary

This progress report outlines the first phase of the SURGE-090 Smart Snake Robot project. The primary goal of this phase is to establish the hardware infrastructure, test single-motor actuation, and implement the software foundation for the 10-degree-of-freedom serpentine locomotion. To date, we have finalized the Dynamixel XL330-M288-T actuator selection, built the initial SolidWorks CAD models for the body segments, and implemented a 50 Hz control loop in Python that supports sensorless obstacle detection through motor current monitoring.

Milestones Completed (25% Phase)

- **Hardware Architecture Definition:** Selected and acquired the ROBOTIS Dynamixel XL330 smart servos, U2D2 serial bridge, and Power Hub Board. Configured 5V power supply to run safely within limits.
- **Mechanical CAD Design:** Modeled segments (Segment v1 & v2) in SolidWorks to support alternating pitch/yaw joint structure, enabling 3D movement.
- **Software HAL & Control Loop:** Written the core control framework in Python, including a modular Hardware Abstraction Layer (HAL) with simulation support, a PD torque controller, and a sinusoidal slither gait engine.
- **Proprioceptive Stall Detection:** Configured 50 Hz current feedback monitoring. Tested and calibrated current threshold spikes (> 350 mA) under manual resistance to trigger obstacle avoidance.

Wokwi ESP32 Sensor Node Simulation

To facilitate untethered head-on obstacle detection, an ESP32 secondary controller was introduced. We have simulated this sensor node in Wokwi, connecting an HC-SR04 ultrasonic distance sensor to the ESP32 dev board. The simulated firmware reads distance data and transmits it via Bluetooth Serial to the main compute unit (Raspberry Pi 4). This architecture isolates real-time high-speed sensing from the primary control loop.

Next Phase Objectives (50% Milestone)

- 3D print the full 10-joint robot frame using eSUN PLA+ filament.
- Assemble the mechanical links and route the TTL daisy-chain cabling.
- Implement the Bluetooth serial receiver on the Raspberry Pi 4 to integrate the ESP32 sensor data.
- Perform initial locomotion tests on high-friction surfaces and calibrate the friction anisotropy pads.